English Medium

EDU TERIA

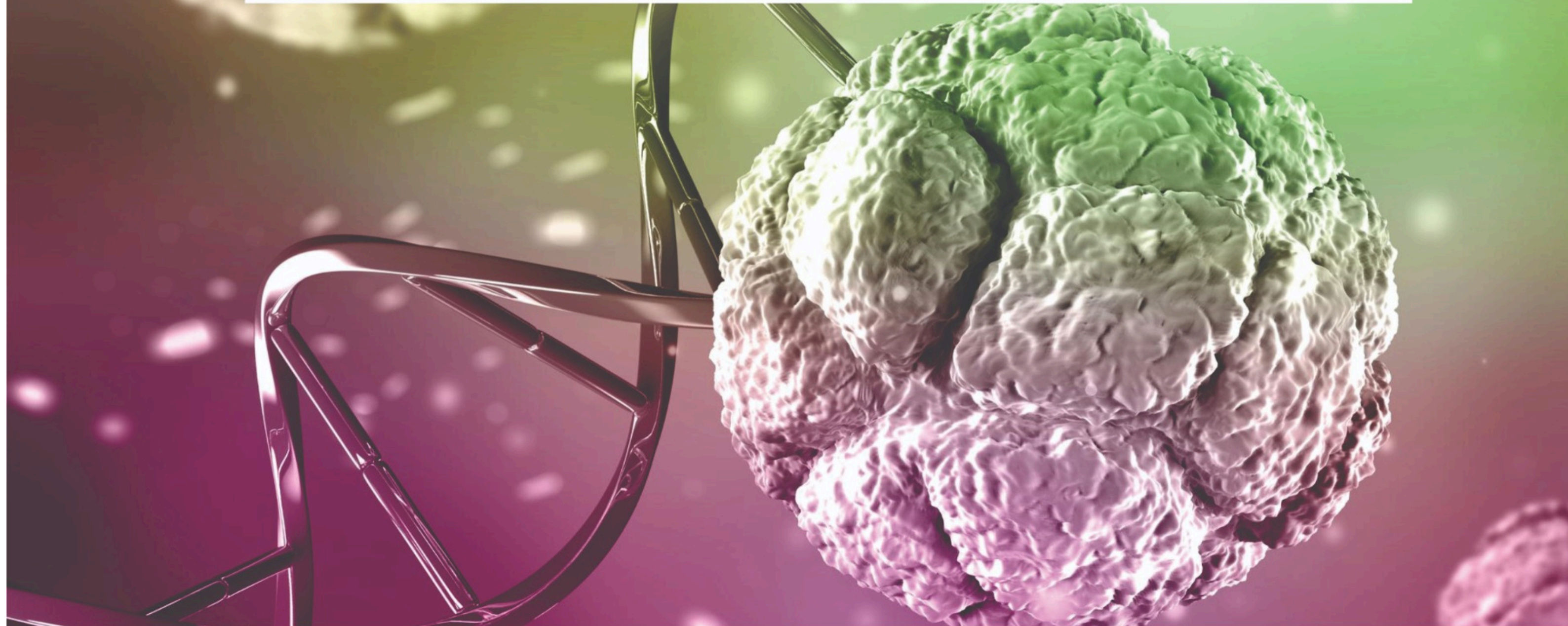
72nd BPSC

Prelims Online Batch

BIOLOGY

LECTURE 4

Nucleus Centrosome, Pollination

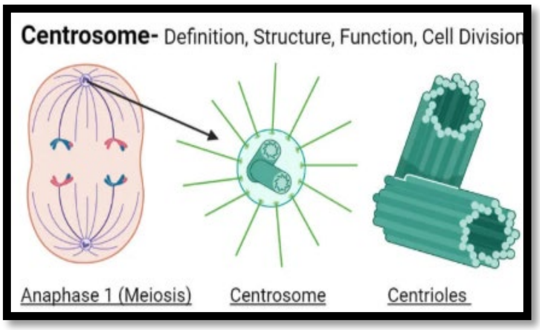


Nucleus Centrosome, Pollination

Centrosome

The centrosome is an important cell organelle found mainly in eukaryotic animal cells. It acts as the main microtubule-organizing center (MTOC) of the cell. The centrosome plays a crucial role during cell division, where it helps in the formation of spindle fibers that separate chromosomes.

- Discovered by Boveri
- Membrane less Cell organelle
- Present in Animal Cell
- Absent in Plant Cell
- The centrosome is a non-membrane bound organelle found exclusively in animal cells.
- **Structure:** It consists of two cylindrical structures known as centrioles.
- **Composition:** These centrioles are surrounded by an amorphous (formless) proteinaceous material called pericentriolar materials.
- **Arrangement:** The two centrioles in a centrosome lie perpendicular to each other. Role in Cell Cycle (Cell Division)
- **Duplication:** During the S-phase of the cell cycle, the centrosome duplicates itself in the cytoplasm.
- **Mitotic Apparatus:** During prophase, the two centrosomes begin to move towards opposite poles of the cell.
- **Spindle Formation:** The centrosomes organize microtubules to form spindle fibers and aster rays. Mitotic Apparatus: During prophase, the two centrosomes begin to move towards opposite poles of the cell. Spindle Formation: The centrosomes organize microtubules to form spindle fibers and aster rays.



Types of Plastids

Plastids are double-membrane cell organelles found in plant cells and algae. They are mainly involved in photosynthesis, storage of food, and synthesis of important substances. Plastids develop from small precursor structures called proplastids.

Plastids are mainly classified into three major types :

Type of Plastid	Main Function	Example
Chloroplast	Photosynthesis (food production)	Green leaves
Chromoplast	Pigment synthesis and storage	Fruits like tomato
Leucoplast	Storage of food materials	Roots and seeds

1. **Chloroplast**

- Green plastids containing chlorophyll pigment.
- Responsible for photosynthesis (production of glucose using sunlight).
- Commonly found in leaf cells and other green parts of plants.
- Structure includes thylakoids, grana, and stroma.

  Chlorophyll— Photosynthetic pigment contains carbon, hydrogen, oxygen, nitrogen and magnesium (only metal)
- Generally, two types of chlorophyll are present in plants—

i. Chlorophyll 'a'	Sensitive for light (photosynthetic active radiation)
ii. Chlorophyll 'b'	

2. **Chromoplast**

- Type of plastid with different types coloring pigments.

- Chromoplast is responsible for multicolor plants :

Parts :—

1. Coloured petals
2. Coloured leaves
3. Coloured fruits

Chromoplast presence in different Plants/Fruits

- Tomato— Lycopene
- Turmeric— Xanthophyll
- Apple— Antholyanin
- Chilly— Cathepsin
- Papaya— Caricaxanthin
- Carrot— Carotin

3. Leucoplast

- Colourless plastids mainly involved in storage of food materials.
- Found in non-photosynthetic tissues such as roots, seeds, and tubers.
- The type of plastid present in that part of plant which is not in direct contact of sunlight.

On the basis of storage of food leucoplastic are of three types:—

i. Amyloplast	ii. Aleuroplast	iii. Elaioplast
↓ Storage of carbohydrate **Eg.**Patato beat, wheat potato rice. [starch storage]	↓ Storage of protein **Eg.** Leguminous plants, pea, soyabean, pulser	↓ Storage of lipid & oil **Eg.** Castor Alsi seeds mustard

Ribosome:

- Discovered by Palade.
- Membrane less cell organelle
- present in prokaryotic as well as eukaryotic Cell.
- Two types of ribosomes are present in prokaryotic and eukaryotic Cell.

i) 70'S' ii) 80's Type

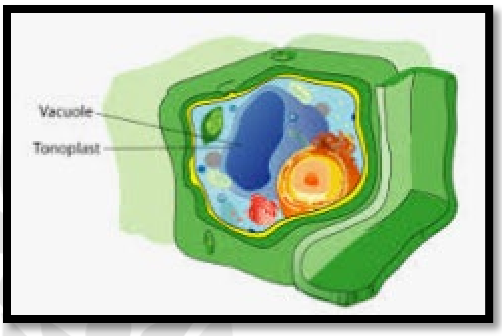
70 ‘s’ = 50 ‘s’ + 30 ‘s’

S— Sedimentation rate or svedverg unit Represent size & density of ribosome

- Ribosome are known as protein factory
- Ribosomes are formed in nucleolus.

Vacuole

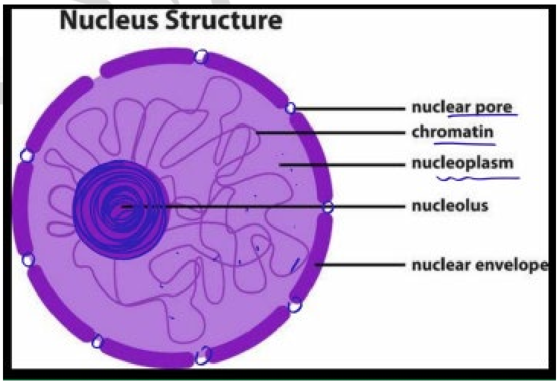
- Single membrane bound structure
- Large size of vacuole is present in plant Cell. (90% part of plant cell)



- Small & temporary vacuoles can be present in Animal Cell.

Nucleus Structure

- The nucleus is a double-membraned, spherical, eukaryotic organelle-first described by Robert Brown--that acts as the cell's control center, housing (1831) DNA in the form of chromatin.
- **Nuclear Envelope:** A double membrane system separating the nucleus from the cytoplasm. The outer membrane is continuous with the Endoplasmic Reticulum (ER).



- The outer membrane is continuous with the Endoplasmic Reticulum (ER).

- **Nuclear Pores:** Minute pores formed by the fusion of the two membranes, controlling the movement of RNA and proteins in/out.
- **Nucleoplasm:** The matrix containing nucleolus and chromatin.
- **Nucleolus:** A non-membrane-bound, spherical structure inside the nucleoplasm, rich in RNA, acting as the site for active ribosomal RNA synthesis. (RTNA +protein) Ribosome)
- **Chromatin:** A network of nucleoprotein fibers (DNA + histone proteins) in interphase cells, which condenses into chromosomes during division.

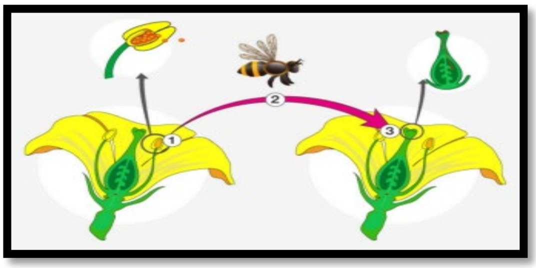
Difference between Plant and Animal Cell

Cell Parts	Plant cell	Animal cell
i) Cell Wall	Present	Absent
ii) Plastid	Present	Absent
iii) Centrosome	Absent	Present
iv) Vacuole	Large size	Small size but temporary
v) Nucleus	Present of the periphery	Present in the Centre

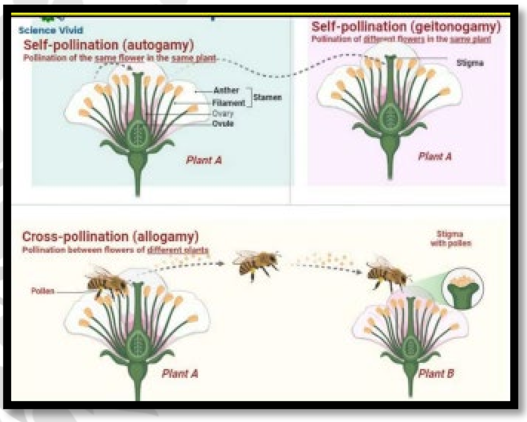
Pollination

- Pollination is the transfer of pollen grains from the anther to the stigma, essential for sexual reproduction in plants, categorized into self-pollination (Autogamy/ Geitonogamy) and cross-pollination (Xenogamy).

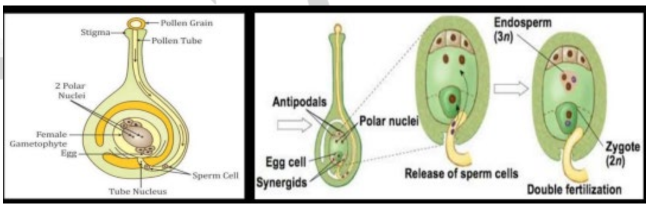
- It requires external agents (wind, water, animals) and relies on compatibility between pollen and stigma (pollen-pistil interaction) to ensure fertilization.



- **Autogamy (Self-pollination):** Transfer of pollen from the anther to the stigma of the same flower. It requires synchrony in pollen release and stigma receptivity.
- **Cleistogamous flowers:** Do not open at all (**e.g.** Viola Oxalis Camelina), ensuring self-pollination



- **Chasmogamous flowers:** Exposed anthers and stigma.
- **Geitonogamy:** Transfer of pollen from the anther to the stigma of another flower on the same plant.
- Functionally cross-pollination (requires agents) but genetically self-pollination.



- **Xenogamy (Cross-p**ollination): Transfer of pollen from the anther to the stigma of a different plant (genetically different).
- This is the only type that brings genetically varied pollen.

Agents of Pollination:

1. **Abiotic Agents**

- **Wind (Anemophily):** Common Pollen is light, non-sticky; stamens are well exposed; stigma is large and feathery.
- **Water (Hydrophily):** Rare (limited to ~30 genera, mostly monocots). Examples: Vallisneria Hydrilla Zostera (sea grasses).

2. **Biotic Agents**

- Animals: Bees, butterflies, birds (hummingbirds), bats, beetles.
- Flowers are large, colorful, fragrant, and rich in nectar.

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